Domain Adaptive Oriented Object Detection From Optical to SAR Images

Hailiang Huang, Jingchao Guo, Huangxing Lin, Yue Huang, *Member, IEEE*, and Xinghao Ding, *Member, IEEE*

***Abstract*— Oriented object detection in synthetic aperture radar (SAR) images presents significant challenges due to the scarcity of labeled data. In contrast, acquiring labeled optical remote sensing images is considerably easier. This article proposes a domain adaptive oriented object detection (DAOOD) model, termed the pixel-instance information transfer-based model (PITM). PITM aims to transfer knowledge from optical to SAR domains, thereby reducing the dependency of oriented SAR object detection on labels. Given the pronounced domain disparity between optical and SAR images, the efficient migration of both visual content and rotating instances is incorporated to bridge the gap in their information distribution simultaneously. Specifically, regarding pixel-level information transfer, speckle noise from SAR images is mixed into the optical domain to form an intermediate domain, thus compensating for the visual difference between the two domains. For instance-level information transfer, considering the angle diversity of rotating objects, multiscale and multidirectional spatial information extraction is combined with decoupled instance-invariant features, enhancing the cross-domain discernment capacity of rotating instances. Experimental results on four DAOOD benchmarks (i.e., two optical datasets to two SAR datasets) demonstrate that the proposed PITM significantly improves oriented object detection performance, even in the absence of labeled SAR images. Specifically, it individually outperforms two source-only models by 88.16% and 54.62% in average precision (AP).**

***Index Terms*— Decoupled instance, Gabor convolutional networks (GCNs), oriented object detection, synthetic aperture radar (SAR), unsupervised domain adaption.**

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I. Introduction

OBJECT detection in synthetic aperture radar (SAR) images has witnessed significant advancements in recent decades, becoming a pivotal method for Earth observations. However, due to its unique observing mode and imaging mechanism, SAR object detection faces two major challenges.

On the one hand, there exist some difficulties in detecting SAR objects due to the unique overhead perspective and long-distance capture. Issues such as significant size disparities among objects, dense arrangements of small objects, and variations in orientation complicate detection efforts. As illustrated in Fig. 1(a), in scenarios where objects are densely and obliquely arranged, the conventional approach of employing multiple horizontal detection boxes often leads to overlapping borders, resulting in excessive coverage of background areas or unintended targets. Similarly, rotating objects, as shown in Fig. 1(c), face the same detection challenges in optical remote sensing images. These issues underscore the inadequacy of horizontal detection boxes in accurately fitting the geometry positions of rotating objects in remote sensing scenes. On the other hand, SAR images are inherently difficult to recognize and interpret, resulting in most SAR images remaining unlabeled and unused. Object detection studies have demonstrated the inseparable link between performance enhancement and substantial labeled data. Consequently, the detection accuracy of unlabeled SAR targets becomes increasing harder to improve, necessitating the efficient information exploration based on unsupervised inference [1].

Oriented object detection and unsupervised domain adaptation offer effective solutions to these two challenges. Specifically, oriented object detection utilizes oriented detection boxes depicted in Fig. 1(b) and (d) to represent rotating objects. This approach provides more comprehensive geometry contour and position information compared to traditional horizontal object detection techniques, thereby improving the accuracy and efficiency of interpreting remote sensing images. In addition, since optical images are easily discernible to the naked eye, labeling them is significantly less costly than labeling SAR images. An unsupervised domain adaptation model is capable of transferring information across domains, bridging the domain gap between a label-rich domain and a new, unlabeled domain. Ideally, a model trained on a labeled optical domain can effectively explore features and recognize objects in the unlabeled SAR target domain. The efficient combination of oriented object detection and

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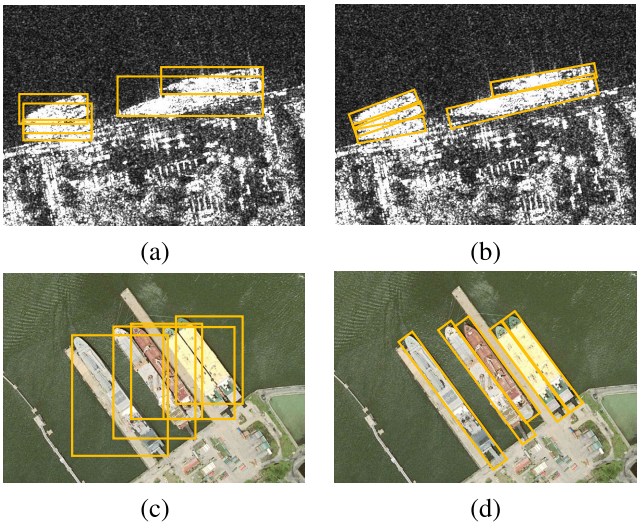


Fig. 1. Illustration of object detection upon rotating objects. (Row 1) Detection results in SAR images using (a) horizontal detection boxes and (b) oriented detection boxes; (row 2) the detection results in optical images using (c) horizontal detection boxes and (d) oriented detection boxes.

unsupervised domain adaptation can significantly enhance the exploration and interpretation reliability of rotating SAR objects.

In recent years, considerable efforts have been devoted to domain adaptive object detection, aiming to mitigate the domain shifts by exploring the fundamental differences between source and target domains. These methods generally fall into three categories, i.e., domain-alignment [2], [3], [4], domain-mapping [5], [6], [7], and self-labeling [8], [9], [10] techniques. However, most of these methods are designed to detect objects in natural scenes. They struggle to effectively capture and transfer information in remote sensing environments, due to the distinct visual content between natural and remote sensing scenes. To address these, some progress has been made in domain adaptive object detection tailed to the remote sensing scenes, including the adaptation task optical-to-optical [11], SAR-to-SAR [12], and optical-to-SAR [13]. These methods effectively capture and transfer domain knowledge by comprehensively considering the characteristics of both optical and SAR domains. However, they typically rely on horizontal detection boxes, which are inadequate for accurately acquiring the geometry location of rotating objects. This limitation renders them unsuitable for oriented object detection in remote sensing contexts. Moreover, the accuracy of existing oriented object detection models [14], [15] heavily depends on the quality of labels, which significantly decreases when labeled data is scarce. To the best of our knowledge, there is currently no work that applies unsupervised domain adaptation to oriented object detection. Therefore, developing a domain adaptive oriented object detection (DAOOD) model, specifically for rotating objects in remote sensing scenes, is both theoretically significant and practically urgent.

With these observations, this article makes the first attempt at DAOOD work from the optical domain to the SAR domain. The detection accuracy of rotating SAR objects depends not only on aligning information distribution across domains but also on precisely capturing the rotating angles. Therefore, the pixel-instance information transfer-based model (PITM) is proposed to explore the domain-specific information together with rotating angles. It achieves the efficient migration of both visual content and rotating instances by leveraging information transfer at pixel and instance levels.

Specifically, in the pixel-level information transfer module, it is obvious to note the distinct visual appearances between optical and SAR images. The speckle noise inherent in the SAR domain is extracted and incorporated into the optical domain, generating an appropriate intermediate domain that contains visual perception elements from both domains. Subsequent adaption procedures are then conducted on this intermediate domain and the target SAR domain. In the instance-level information transfer module, the multiscale and multidirectional feature extraction is initially conducted for the oriented objects varying in both angles and sizes. The Gabor filter is known for depicting the 2-D neural receptive fields in the visual cortex [16], [17]. Its representation is capable of characterizing the oriented frequency structure in images while preserving spatial relation information of objects [18]. To leverage this, the Gabor convolutional network (GCN) [18] is exploited to explore the geometry structures and slant angles of rotating objects, in conjunction with the feature pyramid network (FPN) [19] to capture multiscale and multidirectional features. After this, to enhance the generalization of rotating instances for mitigating domain shifts, rotating instance features are disentangled into domain-invariant and domain-specific representations using common specific decomposition (CSD) [20]. The discriminability of domain-invariant features is further improved by introducing classification compensation losses, thereby strengthening generalization performance. In addition, Gabor information adversarial loss and global instance adversarial loss are computed to refine the consistency of feature distribution between the two domains. Extensive experimental results on four DAOOD benchmarks demonstrate that the proposed PITM accurately detects a greater number of rotating objects in unlabeled SAR images, compared to multiple classic and state-of-the-art domain adaptive object detection models.

The primary contributions of the proposed model are outlined as follows.

1) To the best of our knowledge, this is the first attempt at the DAOOD work. It directly addresses the accuracy limitations of oriented object detection, stemming from the scarcity of labeled SAR training data. A novel PITM is presented by transferring information from both pixel and instance perspectives.
2) The pixel-level information transfer module synthesizes an intermediate domain, mitigating visual discrepancies by incorporating the visual content of both optical and SAR domains.
3) The instance-level information transfer module performs multiscale and multidirectional feature extraction dedicated to rotating objects, enhancing the generalization performance of decoupled rotating instances.

The forthcoming sections of this article are scheduled as follows. Section II provides a review of current research progress. Section III delves into a detailed description of the

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proposed PITM. In Section IV, the experimental results and ablation studies are showcased and analyzed. Section V offers a concise conclusion.

II. Related Works

A. Object Detection for Remote Sensing Images

In recent years, remarkable progress has been made in object detection for remote sensing images, propelled by advancements in the field and the widespread collection of remote sensing data. This trend has led to a surge in research on deep learning-based models, which are known for their efficient detection performance and strong generalization capabilities. For instance, Yao et al. [21] applied the classic object detection model Faster-RCNN [22] to efficiently detect ship objects in remote sensing images. Zhao et al. [23] introduced rotation-invariant deep features driven by channel attention to achieve multiscale object detection in high-resolution remote sensing images. Han et al. [24] designed a multiscale hierarchical channel-spatial collaborative feature enhancement module along with a bidirectional consistency constraint module, to capture global semantic relations and spatial details in remote sensing scenes. However, models that primarily rely on automatically extracted features may struggle to effectively handle complex real-world scenarios, thereby limiting the reliability and interpretability of prediction results. To address this limitation, some studies have sought to integrate image characteristics with deep learning-based object detection models. Zhang et al. [25] presented a feature fusion framework (FEC) that combines attribute scattering center (ASC) features and deep CNN features, to improve the effectiveness of SAR object recognition. Li et al. [26] utilized ASC features to capture local variations in SAR target components, and fused multiscale feature maps to obtain more accurate feature descriptions. Given the complex backgrounds and multiscale object variations in SAR images, Wang et al. [27] built a multifeature fusion network that effectively combines spatial and frequency textural information. Zeng et al. [28] integrated dual-polarization feature representation and the constant false alarm rate to improve the detection sensitivity of small-scale marine objects.

B. Oriented Object Detection for Remote Sensing Images

Research on oriented object detection has been instrumental in addressing the challenges of detecting cluttered and densely packed objects in remote sensing scenes. Ding et al. [29] implemented spatial transformations on regions of interests (ROIs), i.e., converting horizontal ROIs into rotating ROIs to enable oriented object detection. Xie et al. [14] built a comprehensive two-stage oriented detector comprising an oriented region proposal network (RPN) module and an oriented ROIs module. Xu et al. [15] translated the vertices of the rotating border into a relative gliding offset on the horizontal border and introduced an obliquity factor to guide the selection between horizontal or oriented detection. Zheng et al. [30] devised an instance-aware cross-feature fusion module to extract rotating angle information from both spatial and frequency domains. Furthermore, to tackle boundary discontinuity issues in rotating angle regression, Pan et al. [31] proposed a dynamic adjusting label strategy, which fine-tuned labels coded by ground truth to guide angle coding for tendency learning. Yu and Da [32] introduced a novel phase-shifting coder to achieve boundary-discontinuity-free detection in a continuous and differentiable manner. Despite these advancements, research on oriented object detection in SAR images remains relatively limited. Existing approaches predominantly rely on local information perception via CNNs and long-range dependencies captured through Transformer [33], [34]. The efficient detection of oriented SAR targets still requires significant improvement.

C. Domain Adaptive Object Detection for Remote Sensing Images

In response to the challenge posed by limited labels in SAR images, domain adaptive object detection research within the remote sensing field has been undertaken. Huang et al. [35] proposed a transitive transfer method via multisource data to reduce the discrepancies between various domains (including the optical domain and SAR domain) and SAR targets. Shi et al. [12] facilitated knowledge transfer across different SAR domains by incorporating pixel-domain adaptation, multilevel feature domain adaptation, and iterative pseudo-labeling. Building on this foundation, Shi et al. [13] further developed a progressive knowledge transfer approach from the optical domain to the SAR domain. Zhang et al. [36] excavated both global structure and local instance information to align feature representations between SAR and optical images. Yuan et al. [37] applied the CSD module to partition the classifier into domain-common and domain-specific components, improving the former to promote the recognition ability of SAR objects. Luo et al. [38] proposed a semi-supervised cross-domain object detection framework that processes both optical data and limited SAR data, gradually reducing the domain shifts at the image, instance, and feature levels. Zheng et al. [39] presented a dual-teacher framework (i.e., separating into cross-domain and semi-supervised tasks) to address mutual interference between optical and SAR supervision. Han et al. [40] devised a learnable frequency-enhanced feature alignment module to transform features into frequency space. It significantly simplifies the attention solver and effectively captures the domain-specific information.

Despite these advancements, the efforts devoted to domain adaptive object detection in remote sensing images are still in their infancy. Notably, these methods predominantly rely on horizontal detection boxes, compromising the efficiency of detecting rotating targets. To the best of our knowledge, there is currently no domain adaptive object detection method specifically for rotating objects in remote sensing images. Given the scarcity of labeled SAR data, there is an urgent need to develop a DAOOD method tailored for rotating objects.

III. Proposed PITM Method

In DAOOD, it is assumed to avail a source domain $D_s = \{(x_j^s, y_j^s, b_j^s)\}_{j=1}^{N_s}$ consisting of N_s labeled samples, and a target domain $D_t = \{x_k^t\}_{k=1}^{N_t}$ with N_t unlabeled samples. Where x, y, and b represent the images, labels, and bounding boxes annotations, separately. In this context, D_s and D_t specifically

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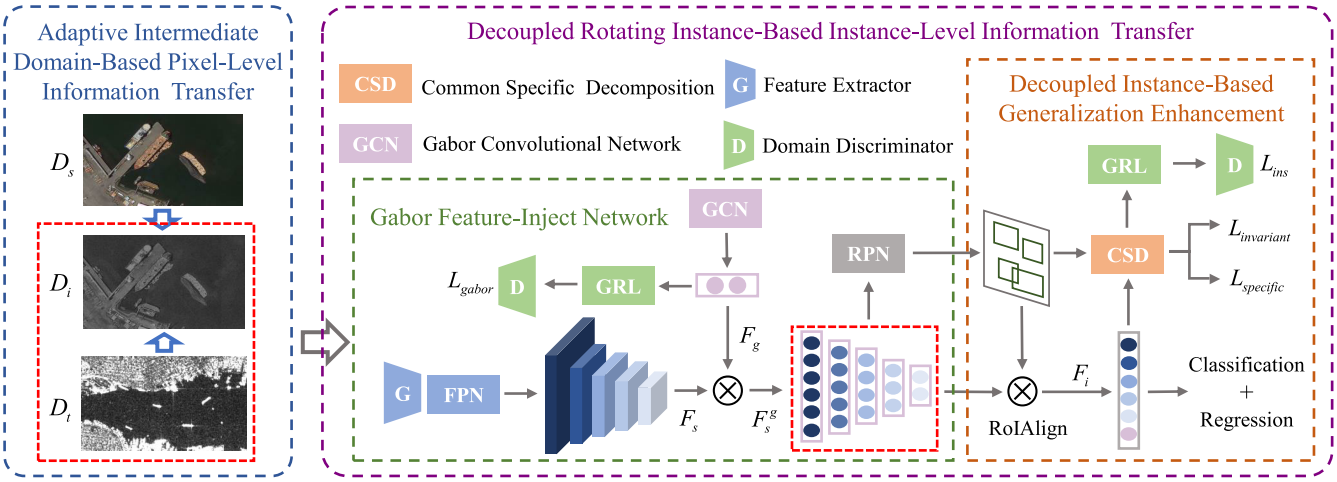


Fig. 2. Framework of the proposed DAOOD model, called the PITM, operates in the following two stages. In the pixel-level information stage, the source domain D_s and target domain D_t are initially combined to synthesize an intermediate domain D_i, which incorporates visual perception elements from both domains. In the instance-level information stage, FPN and GCN are individually exploited to generate multiscale spatial features F_s and Gabor features F_g. The spatial feature exploration is further enriched by injecting F_g, resulting in multiscale and multidirectional representation F_s^g. Following ROIAlign, the rotating instance features F_i are disentangled to generate invariant and specific classification compensation loss (i.e., $L_{\text{invariant}}$ and L_{specific}). In addition, the Gabor information adversarial loss L_{gabor} and global instance adversarial loss L_{ins} are computed to enhance the consistency of feature distribution in the two domains.

refer to optical remote sensing images (i.e., source domain) and SAR images (i.e., target domain). They are sampled from distinct data distributions while sharing an identical label space. The objective is to achieve efficient oriented object detection in the unlabeled D_t by transferring the domain knowledge to mitigate domain disparity.

Generally speaking, the optical and SAR domains have distinct feature distributions due to their different imaging mechanisms. Reliable exploration of rotating objects in SAR images depends on not only the information transfer across domains but also the accurate capture of rotating angles. Thus, the efficient migration of both visual content and rotating instances becomes the central focus of our design. Specifically, a DAOOD model tailored for optical and SAR images is proposed, called the PITM. As depicted in Fig. 2, the framework primarily encompasses the following two stages.

1) *Adaptive Intermediate Domain-Based Pixel-Level Information Transfer:* The speckle noise inherent in the target domain D_t and the fundamental structure information within the source domain D_s are combined to generate the intermediate domain D_i. D_i retains the visual characteristics of both domains, thereby alleviating the visual differences.
2) *Decoupled Rotating Instance-Based Instance-Level Information Transfer:* The FPN and GCN are employed to generate multiscale spatial features F_s and Gabor features F_g. By injecting F_g, the multiscale spatial features are further enriched to explore multiscale and multidirectional content details, resulting in F_s^g. Subsequently, the rotating instance features F_i are disentangled into domain-invariant and domain-specific representations using the CSD module, facilitating the discriminability of rotating instances across domains.

A. Pixel-Level Information Transfer

Optical and SAR domains have distinct object appearances resulting from different imaging mechanisms. Consequently, models trained on optical images struggle to accurately capture geometry contours and precise positions of SAR targets. To bridge the visual gap between source and target domains, deep learning-based style transfer techniques have been widely employed in existing domain adaptive object detection models dedicated to remote sensing images [12], [13], [37]. These methods effectively simulate object appearances while often overlooking information implicit in specific domains, e.g., the correlation of multiple objects deduced from the speckle noise introduced in SAR imaging. To address this, the adaptive intermediate domain synthesis (AIDS) module is designed to realize the pixel-level information transfer, by dynamically capturing the speckle characteristics of the SAR domain. The framework is illustrated in Fig. 3. Specifically, for the jth optical image from D_s (denoted as x_j^s), contrast enhancement is first applied to improve visual effects and highlight object structures in detail. Next, an adaptive speckle background extraction is designed to obtain background images from D_t, considering both small and large speckle noise. Finally, the jth enhanced optical image $x_j^{s'}$ is combined with $x_k^{t'}$, which is produced from an SAR image randomly selected from D_t, generating the jth intermediate image (denoted as x_j^i). The detailed steps are described below.

As a classical method of image contrast enhancement, contrast limited adaptive histogram equalization (CLAHE) [41] calculates the local histogram and redistributes luminance information within a contrast limit parameter, balancing visual content enhancement and noise suppression. It is utilized to characterize the structural details of source images x_j^s. Specifically, x_j^s is divided into subblocks with the size of 8×8, individual computing the equalization by the following transformation:

$$f_u = (U-1) \times \sum_{i=0}^{u} \frac{n_u}{N} \tag{1}$$

where $U = 2^8$, f_u is the pixel value converted from the input pixel with intensity u. n_u is the number of pixels with intensity

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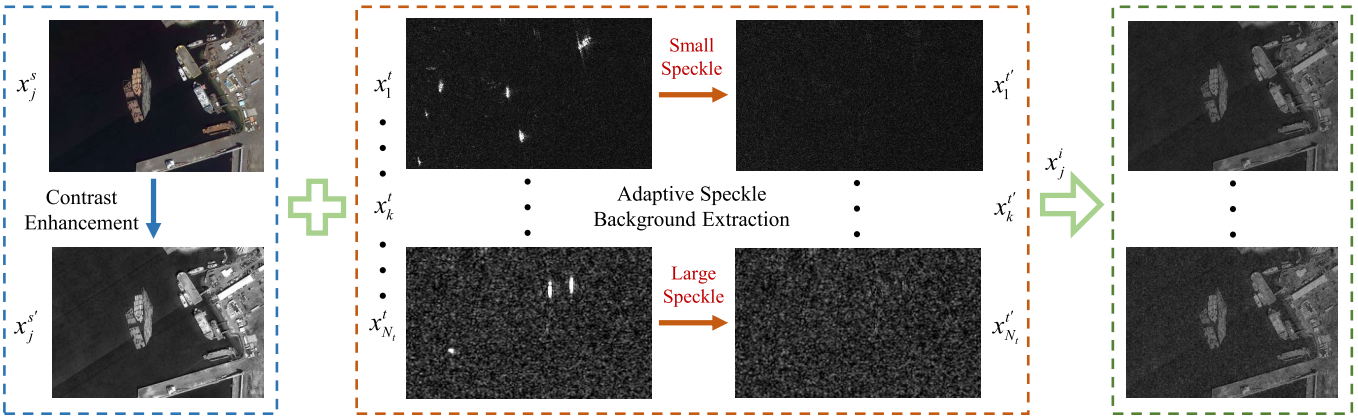


Fig. 3. Illustration of the AIDS. Contrast enhancement is first applied to the jth optical image x_j^s to generate enhanced optical image $x_j^{s'}$. Then, adaptive speckle background extraction is performed to obtain background images (i.e., $x_1^{t'}, \ldots, x_k^{t'}, \ldots, x_{N_t}^{t'}$) from D_t, in consideration of images with both small and large speckle noise. Finally, $x_j^{s'}$ is combined with $x_k^{t'}$ produced from an SAR image randomly selected from D_t, generating the jth intermediate image (i.e., x_j^i).

u, and N is the total number of pixels in each subblock. To prevent noise exaggeration resulting from excessive contrast amplification in smooth regions, the obtained histogram is clipped at the contrast limit. The clipped portions are then equally redistributed back into the histogram bins. Finally, the gray values of all subblocks are combined according to their corresponding positions in the input x_j^s to form the enhanced optical image $x_j^{s'}$. This process improves the visual quality of the input optical images while minimizing noise and artifacts incurred by excessive enhancement in local areas.

Considering the unique characteristics inherent in the speckle noise of SAR images, the main concern of background extraction is how to retain as much speckle information as possible while removing the foreground objects. As depicted in Fig. 3, there exists speckle noise with various sizes in SAR images x_k^t, the binary processing through an adaptive threshold is thus designed to dynamically generate masking images. Specifically, pixel values larger than the threshold δ (i.e., foreground information) are set to 255, and the rest (i.e., background information) are set to 0. For small speckle noise, δ is defined as a perceptible interference threshold η to capture the minute imaging differences. In cases where larger speckle noise significantly surrounds the foreground objects, the threshold determined by pixel value sorting (i.e., γ) is better suited to balance foreground removal and background retention. The adaptive speckle background extraction is conducted as follows:

$$\begin{cases} \delta = \gamma, & \gamma \geq \eta \\ \delta = \eta, & \gamma < \eta \end{cases} \tag{2}$$

where γ is the value of the top 60% of the ranked pixel values from the input x_k^t, and η is experientially set to 70. When $\gamma \geq \eta$, it indicates that 40% of the pixels are higher than the perceptible interference threshold, suggesting a large region with significant speckle noise. In this case, δ is set as γ. Conversely, $\gamma < \eta$ corresponds to x_k^t with small speckle noise, adopting $\delta = \eta$. After obtaining these masking images, they are utilized to detect the contours of foreground objects and characterize them as distorted regions. The background image $x_k^{t'}$ is produced by repairing these regions with adjacent information. Notably, each enhanced optical image $x_j^{s'}$ is combined with a background image generated from a randomly selected x_k^t, resulting in the corresponding intermediate image x_j^i. The subsequent adaption procedures are conducted on this generated D_i and input D_t.

B. Instance-Level Information Transfer

For an efficient migration of rotating instances, the instance-level information transfer is constructed with the Gabor feature-injected network (GFIN) module and the decoupled instance-based generalization enhancement (DIGE) module. These modules achieve multiscale and multidirectional feature extraction and robust recognition of rotating instances, respectively. The detailed information is described in Sections III-B1 and III-B2.

1) Gabor Feature-Injected Network: As aforementioned, the efficient investigation of rotating objects relies on the exploration of multiscale spatial information together with angle information. The comprehensive feature extraction both from multiscale and multidirectional perspectives is thus conducted in our work. Specifically, the multiscale spatial information is first extracted by FPN. The GCN is utilized to extract the Gabor information, which is capable of capturing the geometry structures and angle details. Then, the Gabor information is injected to guide the multiscale spatial feature extraction process, gradually improving detection precision.

The description of the detailed steps is as follows. Due to different gauging distances and imaging mechanisms, the size diversity of detection objects is inevitably existing between optical and SAR domains. To comprehensively capture the geometry contours and position information of both large and small objects, a multiscale exploration of image details is conducted, as shown in Fig. 4(a). First, the multiscale information of the input image is captured by the feature extractor G, with channels of 256, 512, 1024, and 2048, characterizing the objects from coarse to fine levels. Then, these features are integrated into 256 channels by FPN, generating the multiscale spatial features F_s. This aligns with the

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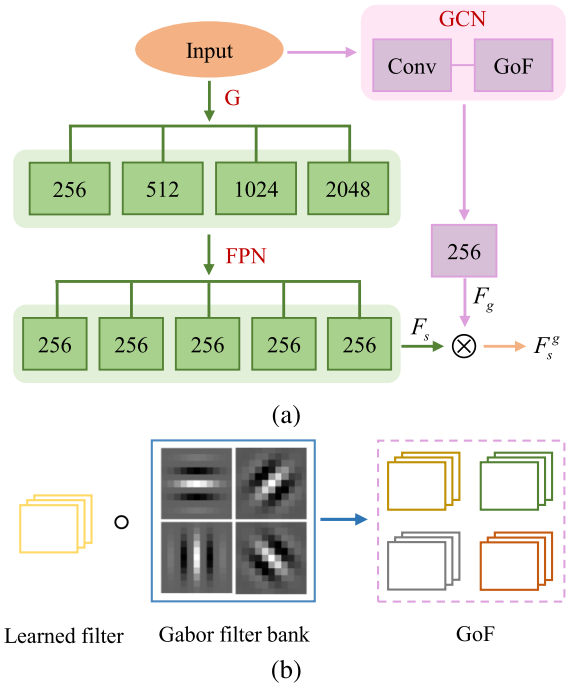


Fig. 4. Illustration of the Gabor feature-inject network (GFIN). (a) Schematic of GFIN, the multiscale information is initially captured by the feature extractor G, with channels of 256, 512, 1024, and 2048. Then, they are integrated into 256 channels by FPN, generating the multiscale spatial features F_s. Upon injection with Gabor features F_g extracted by GCN, F_s evolves into the multiscale and multidirectional representation F_s^g. (b) Composition of GoF, created by Gabor filters oriented in four directions.

multichannel observation mechanism of human eyes, ensuring that the details of objects are captured at various scales.

Moreover, psychophysical and physiological studies have shown that receptive fields in the visual cortex are characterized as being localized, oriented, and bandpass, which can be well depicted by the Gabor filter [42], [43]. The Gabor representation is thus capable of exploring both oriented frequency structures and spatial relations of objects [18]. For the cluttered speckle inherent in input images, the Gabor filter also can capture the speckle information related to local structures while filtering out the irrelevant ones, achieving a definite and comprehensive understanding of visual content. Therefore, the GCN model is employed to guide multiscale spatial feature extraction in this work. Specifically, GCN is a deep-learning network that leverages the Gabor orientation filter (GoF), consisting of learned convolution filters and Gabor filter banks. The Gabor filter is defined as follows:

$$G_o(x, y) = \frac{\|p_o\|^2}{\sigma^2} e^{\frac{-\|p_o\|^2\|(x,y)\|^2}{2\sigma^2}} \left[e^{i p_o(x,y)} - e^{\frac{-\sigma^2}{2}} \right] \tag{3}$$

where $p_o = re^{i\theta_o}$, and $\sigma = 2\pi$ is standard deviation of the Gaussian function. $r = (\pi/2)/\sqrt{2}$ and $\theta_o = o\pi/O$, respectively, represent the frequency and directional information of the filter, with orientation index $o = 0, 1, \ldots, O$.

To incorporate the steerable properties into the GCN, the scale information is encoded in the learned filter C to generate GoF. The definition of the o-th direction filter of the GoF is computed as

$$C_o(x, y) = C \circ G_o(x, y) \tag{4}$$

where $\circ$ represents an element-by-element product operation, naturally generating the GoF constructed by a group of Gabor filters, i.e., $C = (C_1, \ldots, C_o, \ldots, C_O)$, as shown in Fig. 4(b).

Subsequently, the Gabor features F_g is obtained by convolving GoF with the input features F captured by the convolution module, according to

$$F_g = \sum_{o=1}^{O} F \otimes C_o \tag{5}$$

where $O = 4$ is the total number of directions utilized in this work. The produced F_g accesses multidirectional information to enhance the corresponding convolution features F, allowing for a thorough exploration of spatial correlations of objects and specific rotating details. The corresponding four-directional Gabor feature maps are visualized in Fig. 5. From which one can see that the Gabor features can well capture the spatial structures of rotating objects from both optical and SAR images, with different scene complexities. As demonstrated in Fig. 4(a), with the injection of F_g in 256 channels, the multiscale spatial features F_s evolve into F_g^s with multiscale and multidirectional representation.

2) Decoupled Instance-Based Generalization Enhancement: Considering the distinct feature distributions across domains, the efficient exploration and transfer of instance features are crucial to improve the detection accuracy of SAR objects in domain adaptive object detection works [36]. Disentangled instance representation can extract object information with a compact and interpretable structure while being independent of the current observation task [46]. Therefore, we concentrate on capturing instance features invariant across different domains, which depict object characteristics regardless of the domain. In addition, CSD is an efficient domain generalization method that utilizes low-rank decomposition to jointly learn a common component and a domain-specific component [20]. It follows the assumption that classifiers relying on common features better transform to new, unseen domains than those relying on domain-specific features. In this work, CSD is applied to disentangle rotating instance features F_i extracted from RoIAlign [47] into domain-invariant and domain-specific representations. The discernment ability of domain-invariant features is enhanced for the generalization of rotating instances, thereby alleviating their transfer challenges.

Specifically, the CSD works through a domain-invariant classifier Γ_i and domain-specific classifiers $\widehat{\Gamma}_\xi$ of each domain. They separately concentrate on the classification accuracy levering domain-invariant and domain-specific components of rotating instance features, naturally disentangling into domain-invariant and domain-specific representations. Where Γ_i maintains the same parameters across domains, $\widehat{\Gamma}_\xi$ from different domains share Γ_i, with the softmax parameters computed as

$$\widehat{\omega}_\xi = \omega_i + \omega_s \gamma_\xi \tag{6}$$

where $\xi := \{s, t\}$ denote the source domain and target domain, separately. $\omega_i \in \mathbb{R}^{c\times m}$ is the softmax parameter of the domain-invariant classifier, $\omega_s \in \mathbb{R}^{c\times m\times k}$ and $\gamma_\xi \in \mathbb{R}^k$ are the specific domain low-rank parameters.

Noteworthily, the generalization performance of oriented object detection is strongly linked to the discriminability of domain-invariant features. The greater the discriminability of

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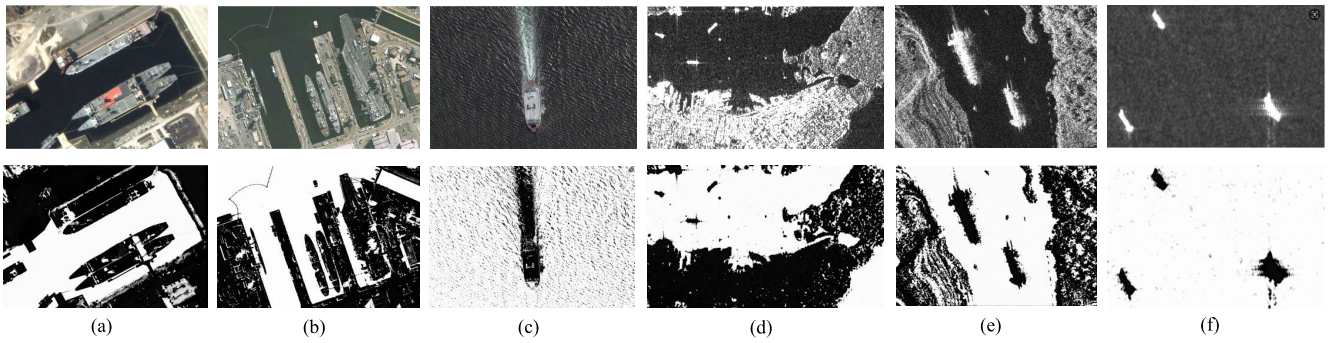


Fig. 5. Illustration of the four-directional Gabor feature maps. (Row 1) (a)–(c) Original optical images selected from the HRSC2016 dataset [44]; (d)–(f) original SAR images selected from the SSDD dataset [45]. (Row 2) Corresponding multidirectional Gabor feature maps of (Row 1).

these features, the more robust the cross-domain object recognition becomes. To achieve this, the invariant classification compensation loss $L_{\text{invariant}}$ and specific classification compensation loss L_{spefic} are designed to enhance the discriminability of domain-invariant features, that is,

$$L_{\text{invariant}} = \mathbb{E}\left(\Gamma_i\left(F_i^s\right), y^s\right) + \mathbb{E}\left(\Gamma_i\left(F_i^t\right), y^t\right)$$
$$L_{\text{specific}} = \mathbb{E}\left(\widehat{\Gamma}_s\left(F_i^s\right), y^s\right) + \mathbb{E}\left(\widehat{\Gamma}_s\left(F_i^t\right), y^t\right)$$

where $\mathbb{E}$ denotes the cross-entropy loss function. F_i^s and F_i^t separately represent the rotating instance features from source and target domains. The comparison labels y^s are directly taken from D_s, while y^t are the pseudo-labels chosen from instances with high confidence.

For each label y^{ξ}, the orthonormality regularizer is utilized to ensure ω_i is orthogonal to ω_s, promoting numerical stability by normalizing each soft parameter into 1. The orthonormality loss L_{ort} is computed according to

$$L_{\text{ort}} = \sum_{l=1}^{L} \left\| I - \Phi(l)^T \Phi(l) \right\|^2 \tag{7}$$

where $\Phi = [\omega_i, \omega_s]$, I represents the identity matrix. $L = 2$ denotes the number of participant domains, i.e., optical domain and SAR domain.

The classification compensation loss calculated from the CSD is derived as

$$L_{\text{csd}} = \alpha_1 L_{\text{invariant}} + \alpha_2 L_{\text{specific}} + \alpha_3 L_{\text{ort}} \tag{8}$$

where α_1, α_2, and α_3 are hyper-parameters to balance the weights of these compensation loss components.

In addition, since adversarial adaptation contributes to enhancing transferability, it is widely employed to minimize domain disparity [3]. The adversarial adaptation with a gradient reversal layer (GRL) [48] is applied to compute the Gabor information adversarial loss L_{gabor} and the global instance adversarial loss L_{ins}, thereby improving the consistency of feature distribution in the two domains. These losses individually achieve feature alignment at pixel and instance levels, with the definition described as follows:

$$L_{\text{gabor}} = -\frac{1}{N_s}\sum_{j=1}^{N_s} \log\left(D_{\text{ga}}\left(F_g^s\right)\right) - \frac{1}{N_t}\sum_{k=1}^{N_t} \log\left(1 - D_{\text{ga}}\left(F_g^t\right)\right)$$
$$L_{\text{ins}} = -\frac{1}{N_s}\sum_{j=1}^{N_s} \log\left(D_{\text{in}}\left(F_i^s\right)\right) - \frac{1}{N_t}\sum_{k=1}^{N_t} \log\left(1 - D_{\text{in}}\left(F_i^t\right)\right) \tag{9}$$

where F_g^{ξ} and F_i^{ξ} represent the Gabor features and instance features obtained from the source domain (namely, s) and the target domain (namely, t), D_{ga} and D_{in} denote the corresponding domain discriminators.

The adversarial loss generated from the combination of L_{gabor} and L_{ins} is defined as

$$L_{\text{adv}} = \beta_1 L_{\text{gabor}} + \beta_2 L_{\text{ins}} \tag{10}$$

where β_1 and β_2 are positive constants to equilibrate these adversarial loss components.

Ultimately, the overall objective function of the proposed PITM model is generated by integrating two detection losses and the above classification compensation loss and adversarial loss, that is,

$$L_{\text{PITM}} = L_{\text{cls}} + L_{\text{reg}} + L_{\text{csd}} + L_{\text{adv}} \tag{11}$$

where L_{cls} and L_{reg} separately denote the classification loss and regression loss. L_{cls} focuses on the classification accuracy of objects, while L_{reg} confirms positioning accuracy by measuring the overlap between the predicted detection boxes and ground-truth bounding boxes.

IV. Experimental Results and Analysis

A. Datasets and Evaluation Criteria

In this section, the adaptation performance of the proposed PITM is evaluated on four benchmark datasets dedicated to oriented object detection, consisting of two optical datasets (i.e., HRSC2016 [44] and ShipRSImageNet [49]) and two SAR datasets (i.e., SSDD [45] and RSDD-SAR [50]), which are described as follows.

1) *HRSC2016 [44]:* This dataset is widely used for ship detection in oriented conditions, featuring image sizes ranging from 300 × 300 to 1500 × 900. It comprises 1061 images gathered from Google Earth, including 70 offshore images containing 90 ship instances and 991 inshore images containing 2886 ship instances. The entire database is adopted as the source domain.
2) *ShipRSImageNet [49]:* This dataset comprises 3435 images originating from various sensors, satellite platforms, locations, and seasons, with the size of 930 × 930. It is divided into 2198 training images, 550 validation images, and 687 testing images. The combination of the training and validation sets is utilized as the source domain.

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3) *SSDD [45]:* This dataset contains 1160 SAR images with 2456 ship instances, divided into a training set of 928 images and a testing set of 232 images. They are acquired from multiple sensors (i.e., RadarSat-2, TerraSAR-X, and Sentinel-1) covering both inshore and offshore scenes, with image sizes ranging from 190 to 668.
4) *RSDD-SAR [50]:* This dataset includes 5000 training images and 2000 testing images, all sized at 512×512. These images originated from Gaofen-3 and TerraSAR-X sensors, encompassing a total of 10263 ship instances captured under various observing modes, polarization modes, and resolutions.

Aiming at the task of detecting a single object category (i.e., ships), four commonly-used evaluation criteria are utilized to assess the accuracy of oriented object detection, measuring the adaptation performance between optical and SAR domains. These criteria include precision rate (P), recall rate (R), average precision (AP), and $F1$-score. Precision and recall, respectively, measure the false alarm rate and missed detection rate according to the following definitions:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{12}$$

where TP indicates the amount of accurately recognized ship targets. FP and FN separately represent the amount of misidentified and missed ship targets. To balance the evaluation performance of precision and recall, there implement AP and $F1$-score, defined as follows:

$$\mathrm{AP} = \int_0^1 P(R) dR \tag{13}$$

$$F1 = \frac{2PR}{P + R}. \tag{14}$$

B. Implementation Details

The proposed model adopts the ResNet-101 [52] architecture pretrained on ImageNet as the basic feature extractor. All experiments are conducted on the PyTorch framework using stochastic gradient descent (SGD) for the training procedure, with a weight decay of 0.0001 and a momentum of 0.9. During the training phase, 24 epochs are applied with an initial learning rate of 0.005, which is divided by 10 at epochs 16 and 22. Each training batch contains two images, respectively, from intermediate and target domains. In the testing phase, the adaptation performance is evaluated by the aforementioned four criteria with an intersection over union (IoU) threshold of 0.5. Moreover, for the adaptation tasks based on source dataset-HRSC2016, the parameters in (8) and (10) are experimentally determined as $\alpha_1 = 0.25$, $\alpha_2 = 0.5$, $\alpha_3 = 0.1$, $\beta_1 = 0.1$, and $\beta_2 = 0.15$. For the adaptation tasks based on source dataset-ShipRSImageNet, these parameters are $\alpha_1 = 0.025$, $\alpha_2 = 0.0375$, $\alpha_3 = 0.01$, $\beta_1 = 0.05$, and $\beta_2 = 0.075$. In addition, the corresponding source codes are taken from the author/website. Since the original parameters of each model are determined according to extensive experiments, reflecting the efficient information exploration of the corresponding visual signals, the parameters are retained. For instance, the source-only models are trained for 36 epochs, following the original setting of the basic oriented detection model [14].

C. Performance Comparison and Analysis

This section demonstrates the performance evaluation of the proposed PITM on four benchmark datasets (i.e., HRSC2016 $\rightarrow$ SSDD/RSDD-SAR and ShipRSImageNet $\rightarrow$ SSDD/RSDD-SAR). To further validate the robustness and efficiency of the PITM, two classic oriented object detection models, oriented RCNN [14] and gliding vertex [15], are utilized as baseline detection models.

1) Overall Performance Analysis: Table I exhibits the adaptation results of the proposed PITM model on four benchmark datasets, based on two basic oriented object detection models. To demonstrate the efficiency of the proposed PITM, the performance is compared with those of source-only models and multiple-domain adaptive object detection models. Specifically, the source-only models refer to those trained on the source dataset (i.e., HRSC2016 and ShipRSImageNet) using oriented RCNN and gliding vertex models. The domain adaptive object detection models evaluated include those designed for natural scenes (i.e., DA-Faster [2], HTCN [3], and SF(DA)^2 [51]) and remote sensing scenes (i.e., OmniAdapt [37]). They detect objects with classic horizontal detection boxes. The unsupervised domain adaptation approaches of these models are applied to two oriented object detection models, detecting the oriented objects in unlabeled SAR images. For a fair comparison, the intermediate images synthesized by the proposed AIDS module are utilized as the training set. In Table I, the best performance among experimental models is highlighted in bold. It is evident that the domain adaptive object detection models show only marginal performance improvement compared to the source-only models. This limited improvement is attributed to the significant differences in the detection boxes and the calculation modes of evaluation, between oriented object detection and classic horizontal object detection. As the first attempt at DAOOD work, the performance demonstration by comparing with source-only models appears more suitable and reasonable.

To explicitly show the efficiency of the proposed PITM, the corresponding comparison results toward source-only models are highlighted in bold, with improvements in red and declines in blue. From which one can see that on four benchmark datasets, most performance criteria exhibit great improvement, except for precision values with a slight decrease. In practical applications, exhaustive detection is often more crucial than precise detection, and enhancing recall performance can result in a tradeoff with precision. In addition, to demonstrate the comprehensive performance improvement over multiple adaptation tasks, their average values are also calculated. Intuitively, the proposed PITM achieves significant performance gains on both basic detection models, i.e., the average AP values by 88.16% for the oriented RCNN model and 54.62% for the gliding vertex model. These results further indicate the robustness and effectiveness of PITM in transferring information across domains.

Moreover, some detection results are visualized in Fig. 6, with the first three rows for the adaptation task HRSC2016 $\rightarrow$ SSDD, and the last three rows for HRSC2016 $\rightarrow$ RSDD-SAR. The images from two SAR datasets originated from different sensors. Considering the diversity of these SAR images, the visualizations focus on objects of varying scene complexities

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TABLE I

RESULTS (%) OF ADAPTATION FROM SOURCE DOMAIN (I.E., HRSC2016 [44] AND SHIPRSIMAGENET [49]) TO TARGET DOMAINS (I.E., SSDD [45] AND RSDD-SAR [50]) ON THE ORIENTED RCNN [14] AND GLIDING VERTEX [15] MODELS

	Method	Oriented RCNN [14]				Gliding Vertex [15]			
		AP	Recall	Precision	F1-score	AP	Recall	Precision	F1-score
HRSC2016 → SSDD	Source-only	20.31	25.46	**28.50**	26.89	22.00	29.30	54.42	38.09
	DA-Faster [2]	25.12	34.04	10.18	15.67	29.22	31.14	**62.27**	41.52
	HTCN [3]	29.04	38.64	10.53	16.55	23.22	30.94	47.78	37.56
	SF(DA)^2 [51]	26.36	31.14	18.24	23.01	31.69	40.66	46.06	43.19
	OmniAdapt [37]	24.40	30.77	12.52	17.80	28.30	33.70	53.80	41.44
	PITM [Ours]	**37.60**	**44.69**	26.44	**33.22**	**42.22**	**50.73**	45.04	**47.72**
		85.13%↑	75.53%↑	7.23%↓	23.54%↑	90.91%↑	73.14%↑	17.24%↓	25.28%↑
ShipRSImageNet → SSDD	Source-only	21.59	30.04	**28.69**	29.35	19.87	31.87	**21.67**	25.80
	DA-Faster [2]	29.70	47.80	23.37	31.39	22.54	41.76	16.75	23.91
	HTCN [3]	27.42	42.49	12.69	19.54	21.13	34.43	18.25	23.86
	SF(DA)^2 [51]	29.18	44.51	21.81	29.28	25.00	41.94	19.88	26.97
	OmniAdapt [37]	30.36	42.67	8.79	14.58	26.65	47.62	19.98	28.15
	PITM [Ours]	**38.58**	**57.51**	20.01	**29.69**	**29.26**	**52.20**	20.08	**29.00**
		78.69%↑	91.44%↑	30.25%↓	1.16%↑	47.26%↑	63.79%↑	7.34%↓	12.80%↑
HRSC2016→RSDD-SAR	Source-only	9.09	9.86	**21.24**	13.47	12.95	15.47	24.56	18.98
	DA-Faster [2]	10.30	12.80	11.73	12.24	14.11	13.87	39.90	20.58
	HTCN [3]	12.18	13.97	12.85	13.39	13.77	11.06	40.03	17.33
	SF(DA)^2 [51]	13.29	15.24	14.32	14.77	13.40	11.66	34.75	17.46
	OmniAdapt [37]	12.82	13.74	9.80	11.44	15.28	20.32	28.19	23.62
	PITM [Ours]	**19.69**	**26.67**	11.78	**16.34**	**20.31**	**22.53**	**48.01**	**30.67**
		116.61%↑	170.49%↑	44.54%↓	21.31%↑	56.83%↑	45.64%↑	95.48%↑	61.59%↑
ShipRSImageNet → RSDD-SAR	Source-only	22.35	29.61	13.98	18.99	23.79	36.57	19.36	25.32
	DA-Faster [2]	23.24	34.59	18.07	23.74	28.82	40.84	18.88	25.82
	HTCN [3]	26.75	38.00	**21.30**	**27.30**	26.96	38.24	18.40	24.85
	SF(DA)^2 [51]	29.29	37.83	12.51	18.80	25.34	38.70	**24.05**	**29.66**
	OmniAdapt [37]	31.38	42.48	13.69	20.71	24.40	37.23	18.27	24.51
	PITM [Ours]	**38.49**	**45.96**	14.91	22.52	**29.37**	**44.35**	18.05	25.66
		72.21%↑	55.22%↑	6.65%↑	18.59%↑	23.46%↑	21.27%↑	6.77%↓	1.34%↑
Average Performance Improvement		88.16%↑	98.17%↑	18.84%↓	18.55%↑	54.62%↑	50.96%↑	16.03%↑	24.59%↑

and sizes, including large objects in inshore scenes, small objects in inshore scenes, and small objects in offshore scenes. The images with ground truth, detection results based on source-only, and the proposed PITM models are depicted from left to right. It can be observed that PITM captures more ship objects and predicts more accurate location information compared with source-only models. The efficient performance under different types of SAR images further indicates the robustness and adaptability of PITM. In addition, note that under the same experiment condition, detection accuracy is inevitably influenced by the complexity of scenes, i.e., appearing missed or falsely detected around the confusing ocean clutter and complex coastlines. Hence, it is vital for object detection to capture the foreground objects while removing interference from background information. Future work will be devoted to the exploration and usage of saliency information, on the basis of the proposed PITM model. In summary, these results and those in Table I indicate that the proposed PITM significantly improves the performance of oriented object detection without using labeled SAR images, emphasizing its theoretical and practical value.

2) Ablation Study of Different Components: To assess the effectiveness of each component of the proposed PITM, a series of ablation studies are conducted on the adaptation task from the HRSC2016 dataset to the SSDD dataset, as shown in Table II. For a clear comparison, the best results for each evaluation index (i.e., AP, recall, precision, and $F1$-score) are highlighted in bold. In source-only situations, the performance of basic oriented detection models (i.e., oriented RCNN and gliding vertex) trained on intermediate images synthesized by the AIDS module is better than those trained on the original source images. This proves the effectiveness of the AIDS module in reducing the visual disparity between optical and SAR domains. In the DAOOD mode, the performance of various component combinations is evaluated. Performance drops when any one module is removed, indicating that domain shifts are gradually alleviated by integrating different modules. The proposed PITM model maximizes domain information transfer, further implying the complementarity of each module and the rationality of the model design.

3) Performance Comparison on Intermediate Domain Synthesis Methods: In this section, we further present the performance comparison of the model using different intermediate images, based on the adaptation task HRSC2016 → SSDD. Table III displays the results trained on original source images and the intermediate images synthesized by the proposed AIDS and three classic and latest style transfer models (i.e., AdaIN [53], CycleGAN [54], and CAP-VSTNet [55]). The best results of each evaluation index are clearly shown in bold. It is evident that the performance yielded by style

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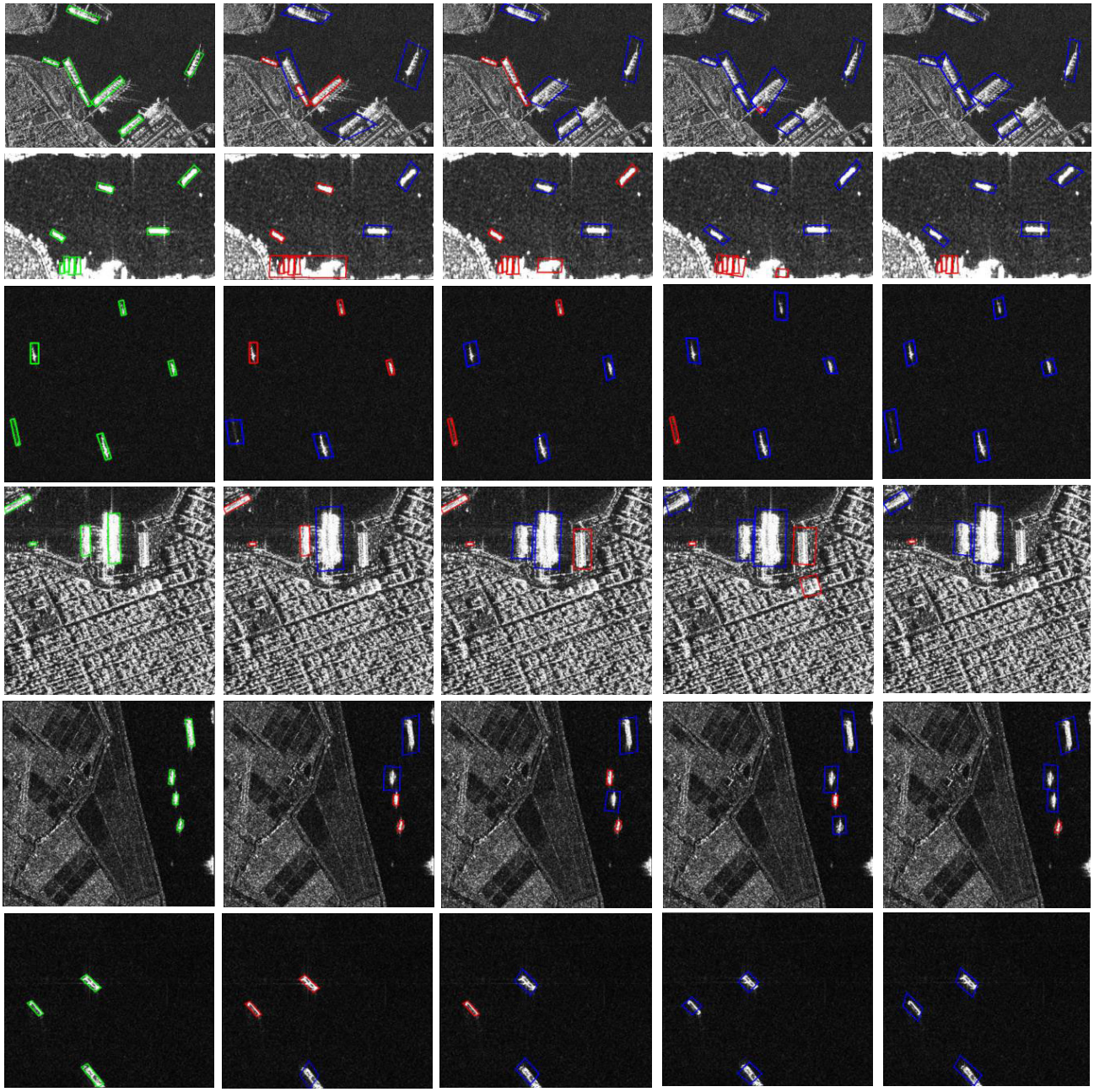


Fig. 6. Illustration of detection results. (Rows 1–3) (a) Ground truth image selected from the SSDD dataset, and the adaptive detection results of: (b) oriented RCNN with source-only training dataset; (c) gliding vertex with source-only training dataset; (d) proposed PITM based on oriented RCNN; and (e) proposed PITM based on gliding vertex. (Rows 4–6) Corresponding adaptive detection results from the RSDD-SAR dataset. Where the green and blue boxes represent the ground truth and accurate detection results, and the red boxes denote the objects that are missed or falsely detected.

transfer models is roughly close to that obtained from original optical images, indicating the limitation of domain information transfer using these generated intermediate images. In contrast, the performance using the proposed AIDS shows significant improvement on both the oriented RCNN and gliding vertex models. This is because the intermediate images produced by AIDS incorporate the object structures of the optical domain and the speckle characteristics of the SAR domain. The synthesized content retains the visual perception results of both domains, significantly alleviating their visual differences.

In addition, to intuitively observe the results of the adaptation task, the training images and the corresponding instance feature representations are visualized in Fig. 7, based on the oriented RCNN model. The original optical images and four generated intermediate images are shown in the first row, displaying the remission degree of visual differences. In the second row, the corresponding instance features are visualized by t-distributed stochastic neighbor embedding (t-SNE) [56]. The feature points generated from optical images, intermediate images, and SAR images are, respectively, depicted as the

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TABLE II
ABLATION RESULTS (%) OF ADAPTATION ON HRSC2016 → SSDD LEVERAGING THE ORIENTED RCNN [14] AND GLIDING VERTEX [15] MODELS

Method		Oriented RCNN [14]				Gliding Vertex [15]			
		AP	Recall	Precision	F1-score	AP	Recall	Precision	F1-score
Source-only	Original	20.31	25.46	28.50	26.89	22.00	29.30	54.42	38.91
	AIDS	22.58	30.40	16.86	21.69	30.21	35.90	**59.94**	44.90
DAOOD	AIDS+GFIN	23.32	30.77	18.77	23.32	29.97	34.25	64.26	44.68
	AIDS+DIGE	28.90	41.39	17.53	24.63	33.11	43.77	40.03	41.82
	GFIN+DIGE (w/o $Loss_{adv}$)	28.11	32.05	**45.45**	**37.59**	23.96	30.22	42.09	35.18
	GFIN+DIGE	26.47	34.07	12.77	18.58	21.05	29.85	52.41	38.04
	AIDS+GFIN+DIGE (w/o $Loss_{adv}$)	33.93	42.31	20.68	27.78	31.68	46.70	32.29	42.68
	AIDS+GFIN+DIGE (i.e., PITM)	**37.60**	**44.69**	26.44	33.22	**42.22**	**50.73**	45.04	**47.72**

TABLE III
PERFORMANCE COMPARISON (%) OF ADAPTATION ON HRSC2016 → SSDD USING DIFFERENT INTERMEDIATE DOMAINS

	Method	AP	Recall	Precision	F1-score
Oriented RCNN [14]	Original	26.47	34.07	12.77	18.58
	AdaIN [53]	25.64	37.73	13.15	19.50
	CycleGAN [54]	22.21	28.75	18.00	22.14
	CAP-VSTNet [55]	23.27	29.49	17.22	21.74
	AIDS [Ours]	**37.60**	**44.69**	**26.44**	**33.22**
Gliding Vertex [15]	Original	21.05	29.85	52.41	38.04
	AdaIN [53]	21.52	33.15	24.30	28.04
	CycleGAN [54]	26.31	32.42	45.38	37.80
	CAP-VSTNet [55]	30.46	31.32	**57.00**	40.43
	AIDS [Ours]	**42.22**	**50.73**	45.04	**47.72**

TABLE IV
PERFORMANCE (%) WITH/WITHOUT ADAPTIVE THRESHOLD

	Criteria	Oriented RCNN [14]			Gliding Vertex [15]		
		$\delta = \eta$	$\delta = \gamma$	PITM	$\delta = \eta$	$\delta = \gamma$	PITM
HRSC2016 → SSDD	AP	29.93	23.99	**37.60**	27.61	29.69	**42.22**
	Recall	38.46	33.88	**44.69**	37.91	44.32	**50.73**
	Precision	27.67	**27.99**	26.44	31.27	23.50	**45.04**
	F1-score	32.18	30.65	**33.22**	34.27	30.71	**47.72**
HRSC2016 →RSDD-SAR	AP	11.50	13.48	**19.69**	16.77	15.04	**20.31**
	Recall	13.80	14.91	**26.67**	**24.77**	24.67	22.53
	Precision	10.94	8.25	**11.18**	30.84	28.78	**48.01**
	F1-score	12.20	10.62	**16.34**	27.47	26.57	**30.67**

colors of red, orange, and green. Some intriguing observations can be obtained: 1) these presented intermediate images successfully capture the appearance characteristics of SAR images while preserving the basic contours of objects from the optical images; 2) the background noise simulated by style transfer models is arranged randomly and messily, ignoring the information inherent in speckle noise; and 3) under the premise of proposed GFIN and DIGE modules, there is an amount of overlap in the distribution of the instance features from different domains, as seen in the t-SNE results in Fig. 7(a), while the proposed AIDS module significantly reduces the domain gap, reflected in the feature points uniformly fitting together. The t-SNE results are in accord with the findings presented in Table III, further indicating that the proposed PITM achieves a more reliable understanding of image content and a more accurate detection of rotating objects.

4) Impact of Adaptive Threshold in AIDS Module: In this section, we further analyze the effects of the adaptive threshold in the proposed AIDS module. For that, the performances yielded from the proposed PITM with/without the adaptive threshold are investigated on two DAOOD benchmarks, as shown in Table IV. For the former case, the adaptive threshold denoted in (2) is utilized to dynamically generate the background images. On the contrary, fixed values (i.e., $\delta = \eta$ and $\delta = \gamma$) are used for the case of without adaptive threshold. They represent the AIDS module that uses a threshold solely determined by the perceptible interference threshold η and that uses a threshold only decided by pixel value sorting γ, separately.

From Table IV, it is evident that the performance of the proposed PITM is influenced by the choice of threshold, reflecting the varying degrees of visual migration induced by the generated intermediate images. The proposed model with the adaptive threshold (i.e., PITM) consistently outperforms those without it. This finding indicates the effectiveness of its generated intermediate images, which successfully balance the extraction of speckle information and the removal of foreground objects. The adaptive threshold in AIDS significantly contributes to pixel-level information transfer. However, it is important to note that the dynamical extraction of speckle noise relies on adaptive binary processing within the AIDS module, making it challenging to fully retain the implicit physical characteristics of SAR images. Future work will focus on exploring and extracting these inherent physical characteristics to further improve pixel-level information transfer.

5) Impact of Directions of Gabor Filter: To examine the impact of direction, we investigate the performance of the proposed PITM using the Gabor features yielded from different total number of directions O, ranging from $O = 1$ to $O = 8$. Where $O = 1$ corresponds to the Gabor features focusing primarily on the geometry structures of objects instead of the angle details. The corresponding results on adaptation task HRSC → SSDD are shown in Table V, using oriented RCNN and gliding vertex models. The top-three performances in terms of each evaluation criteria are highlighted in bold, respectively, marked in red, blue, and black. It reveals that the adaptation performance varies with the total number of directions, indicating that Gabor features from different directions efficiently explore distinct information. Notably, the PITM with Gabor features from four directions (i.e., $O = 4$)

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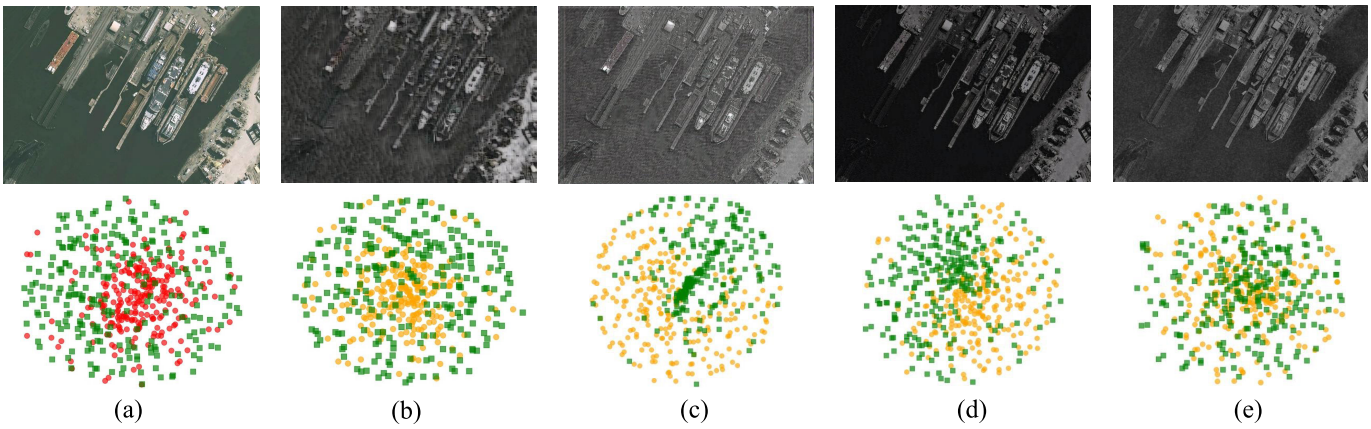


Fig. 7. Illustration of input training images and t-SNE results of the instance feature distributions. (Row 1) (a) Original source image chosen from the HRSC2016 dataset [44], and the intermediate images generated by: (b) AdaIN [53]; (c) CycleGAN [54]; (d) CAP-VSTNet [55]; and (e) proposed AIDS module. (Row 2) Corresponding t-SNE results, where the red and orange round points represent the instance features of source and intermediate images, and the green square points denote the instance features of target images.

TABLE V

PERFORMANCE (%) OF ADAPTATION ON HRSC2016 → SSDD USING DIFFERENT TOTAL NUMBER OF DIRECTIONS O

	Criteria	1	2	3	4	5	6	7	8
Oriented RCNN [14]	AP	**36.97**	32.29	36.72	**37.60**	25.03	24.83	**37.11**	35.83
	Recall	42.12	38.46	**44.32**	**44.69**	34.43	28.57	43.04	**45.44**
	Precision	**24.68**	14.27	16.91	**26.44**	14.48	8.39	**30.92**	17.23
	F1-score	**31.12**	20.82	24.48	**33.22**	20.39	12.97	**35.99**	24.98
Gliding Vertex [15]	AP	32.98	27.84	32.37	**42.22**	32.67	**35.16**	26.63	**33.55**
	Recall	**42.12**	32.78	41.21	**50.73**	40.29	**45.05**	30.95	41.03
	Precision	40.93	**53.43**	35.05	**45.04**	30.51	38.68	**50.90**	40.58
	F1-score	**41.52**	40.63	27.88	**47.72**	34.72	**41.62**	38.49	40.80

obtains the most top-three performances, making $O = 4$ the optimal direction index, as utilized in (5). The proposed PITM achieves the superior performance in oriented object detection of unlabeled SAR images, by efficiently capturing the angle information.

6) Impact of IoU Threshold: IoU is an essential parameter in evaluating object detection performance, presenting the ratio of overlapping areas between the predicted detection boxes and the ground truth bounding boxes. An object is considered accurately recognized when its IoU values exceed a predefined threshold. Consequently, a higher threshold implies the more stringent detection criteria. To analyze performance variation due to different IoU thresholds, we present a comparative study, as demonstrated in Fig. 8. This includes the AP results of the proposed PITM and source-only models, based on the oriented RCNN and gliding vertex models. Clearly, the trend of performance variation is consistent across the two transfer tasks, showing a decrease in performance as the IoU threshold increases. In addition, the AP values of the proposed PITM consistently outperform those of the source-only models, indicating higher precision and robustness in bounding box regression. These findings underscore the effectiveness of PITM in handling the domain adaptation challenges between optical and SAR images.

7) Computational Complexity Analysis: In addition to detection accuracy, computational complexity is an indispensable aspect of the practical application of the

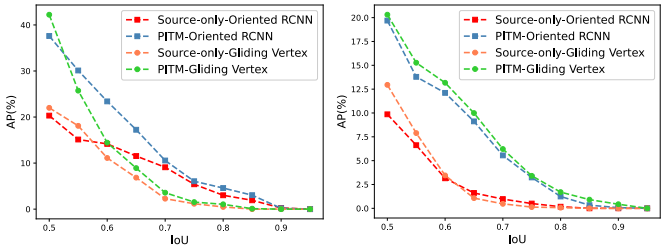


Fig. 8. Performance with the variation of IoU thresholds on adaptation task: (a) HRSC2016 → SSDD and (b) HRSC2016 → RSDD-SAR.

TABLE VI

COMPUTATIONAL COMPLEXITY COMPARISON OF DIFFERENT MODELS BASED ON ORIENTED RCNN MODEL [14]

Method	FLOPs (G)	Parameters (M)	Per Time (ms)
Source-only	240.10	60.12	128.78
DA-Faster [2]	259.82	112.00	141.12
HTCN [3]	279.83	115.69	143.68
SF(DA)^2 [51]	240.10	60.64	132.31
OmniAdapt [37]	344.44	141.30	161.92
PITM [Ours]	243.70	61.74	133.73

DAOOD model. For that, we comprehensively analyze the computational complexity of each model under comparison. These models are operated based on the oriented RCNN model, for the adaptation task ShipRSImageNet → SSDD. The corresponding results are demonstrated in Table VI, the metrics include the floating-point operations per second (FLOPs), the number of model parameters, and the average running times per SAR image. All experiments were conducted in a unified platform equipped with a single NVIDIA RTX 3080 Ti. The results indicate that the proposed PITM gains little computational complexity toward the source-only model. Although the SF(DA)^2 demonstrates slightly lower computational complexity than the PITM, its adaptation performance is significantly inferior to ours, as they cannot accurately describe the domain information of the rotating objects. In contrast, the PITM excels in detecting rotating objects in unlabeled SAR images while maintaining

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acceptable computational demands, making it a competitive option in practice.

V. Conclusion

In this article, we propose a DAOOD model to mitigate domain disparity between a labeled optical domain and an unlabeled SAR domain, namely, the PITM. To comprehensively transfer detection abilities across domains, the efficient migration of both visual content and rotating instances is employed to bridge the gap in information distribution between optical and SAR domains. Specifically, the proposed PITM presents two stages: 1) pixel-level information transfer module captures the speckle inherent in the SAR domain to generate an intermediate domain, significantly mitigating the visual disparity between the source and target domains; and 2) instance-level information transfer module enhances the multiscale and multidirectional spatial feature extraction for rotating objects and improves generalization of rotating instances across domains. Experimental results on four DAOOD benchmarks demonstrate that the proposed PITM outperforms multiple classic and state-of-the-art domain adaptive object detection models in oriented object detection of unlabeled SAR images.

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